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Google Cloud

Start with a solid platform

Module 2 - User interface
Google Cloud Computing Foundations

Course map

01 — So, what's the cloud anyway?

02 — Start with a solid platform

03 — Use Google Cloud to build your apps

04 — Where do I store this stuff?

05 — There's an API for that!

06 — You can't secure the cloud, right?

07 — It helps to network

08 — Keeping an eye on things

09 — You have the data, but what are you doing with it?

10 — Let machines do the work

Google Cloud skill badges

Objectives

01 Explore the Google Cloud console.

02 Examine how projects are the basis for enabling and using Google Cloud services.

03 Identify how billing works in Google Cloud.

04 Install and configure the Cloud SDK.

05 Recognize the different use cases for using Cloud Shell and the Cloud Shell code editor.

06 Explore how APIs work.

07 Manage Google Cloud services from a mobile device.

Agenda

01 The Google Cloud console

02 Understanding projects

03 Google Cloud billing

04 Install and configure the Cloud SDK

05 Cloud Shell

06 Lab: A Tour of Google Cloud Hands-on Labs

07 Lab: Getting Started with Cloud Shell and gcloud

08 Google Cloud APIs

09 The Cloud Mobile App

10 Quiz

11 Summary

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You can interact with Google Cloud in four ways

01

Google Cloud console

02

Cloud SDK and Cloud Shell

03

APIs

04

Cloud Mobile App

You can interact with Google Cloud in four ways

01

Google Cloud console

02

Cloud SDK and Cloud Shell

03

APIs

04

Cloud Mobile App

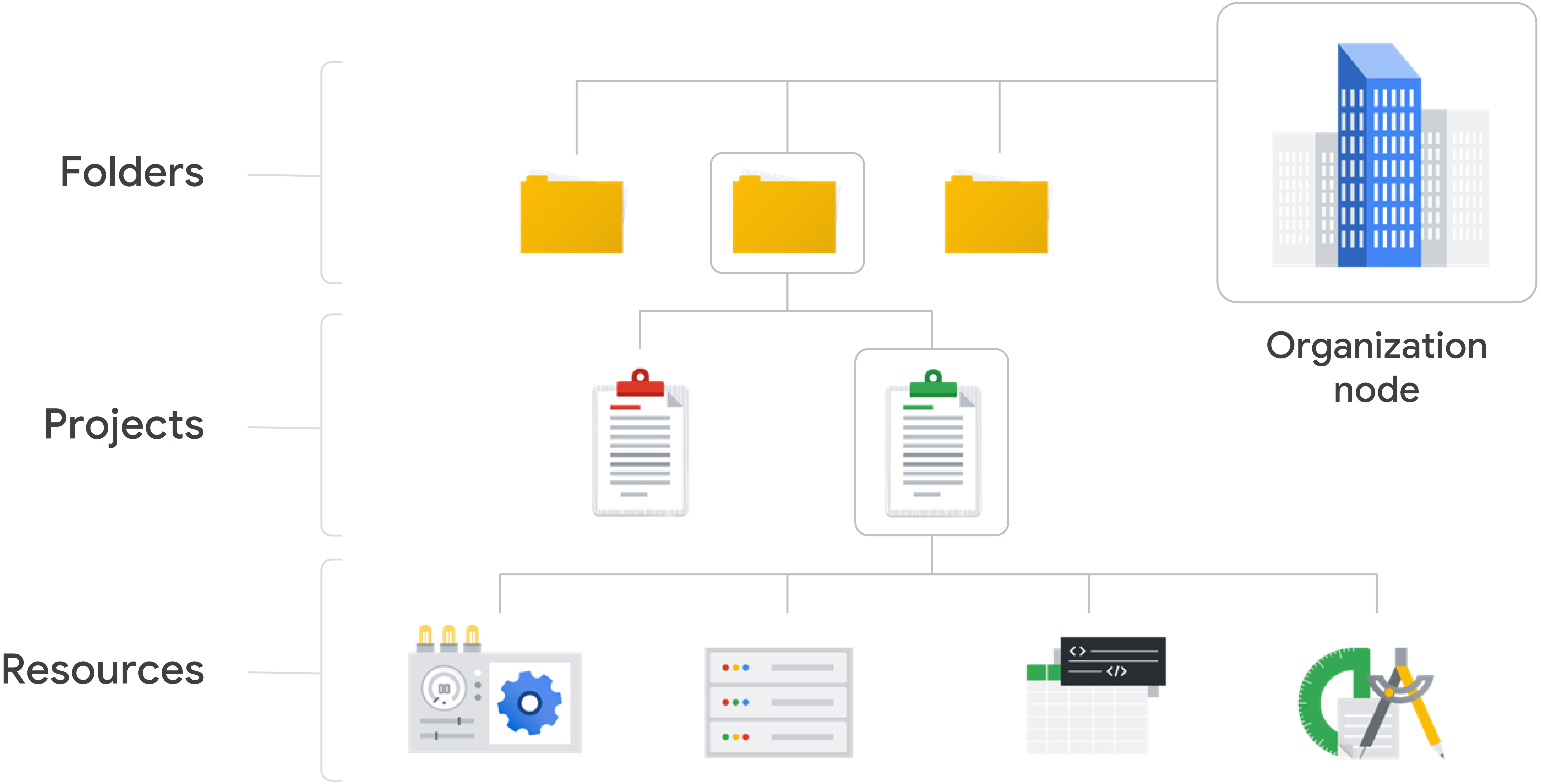
The Google Cloud console provides web-based interaction

console.cloud.google.com

- Simple web-based graphical user interface
- Easily find resources, check their health, have full management control over them, and set budgets
- Provides a search facility to quickly find resources and connect to instances via SSH in the browser

Agenda

Resources are hierarchical



Projects are the basis for utilizing Cloud services

01 Projects are separate entities under the Organization node

02 Projects hold resources, each of which belongs to just one Project

03 Projects can have different owners and users

04 Projects are billed separately

Project attributes vary in uniqueness and immutability

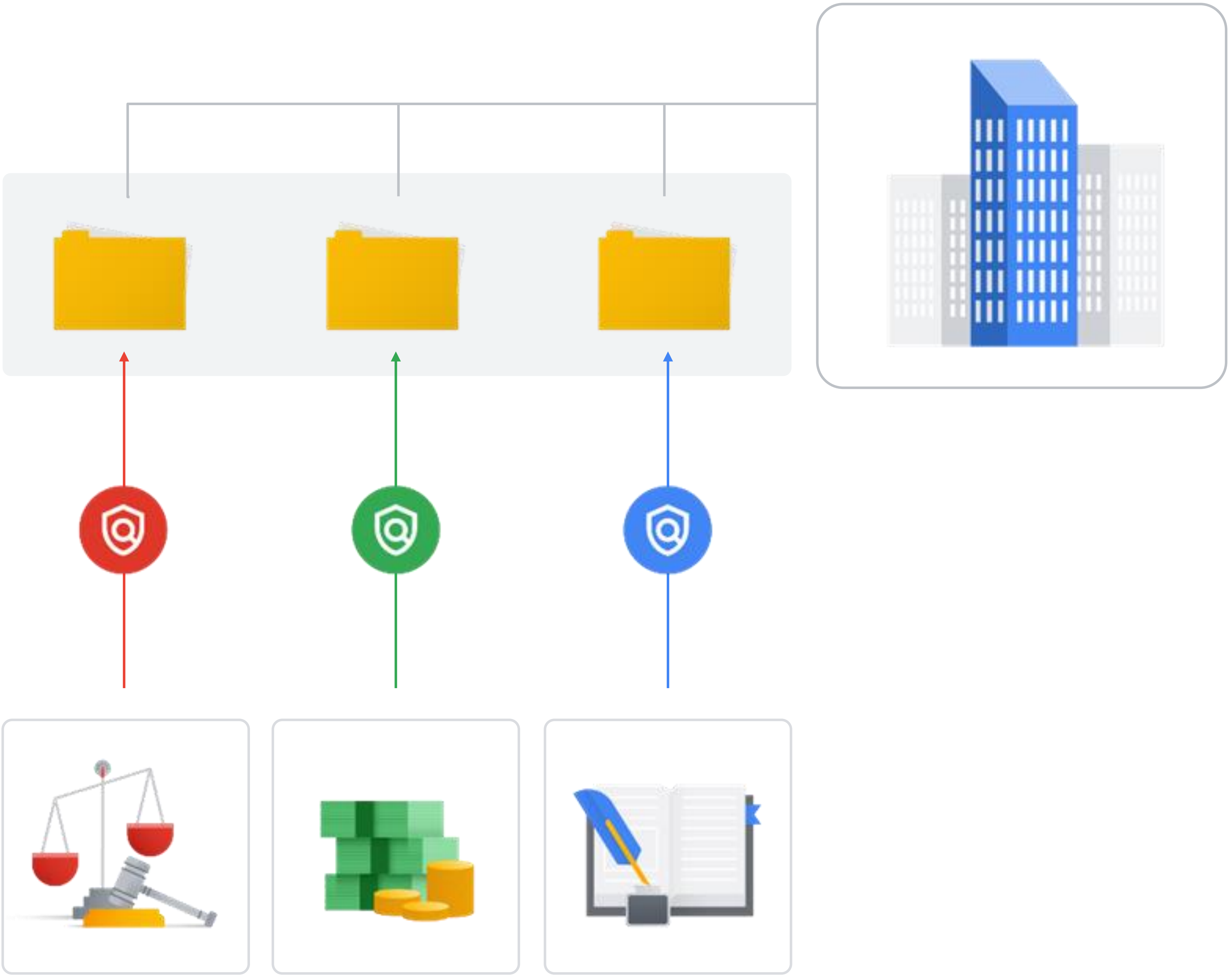
Project ID	Project name	Project number
Globally unique	Need not be unique	Globally unique
Assigned by Google Cloud but mutable during creation	Chosen by you	Assigned by Google Cloud
Immutable after creation	Mutable	Immutable

Resource Manager manages Projects

Resource Manager tool

- Gather a list of projects
- Create new projects
- Update existing projects
- Delete projects
- Recover previously deleted projects
- Access through RPC API and REST API

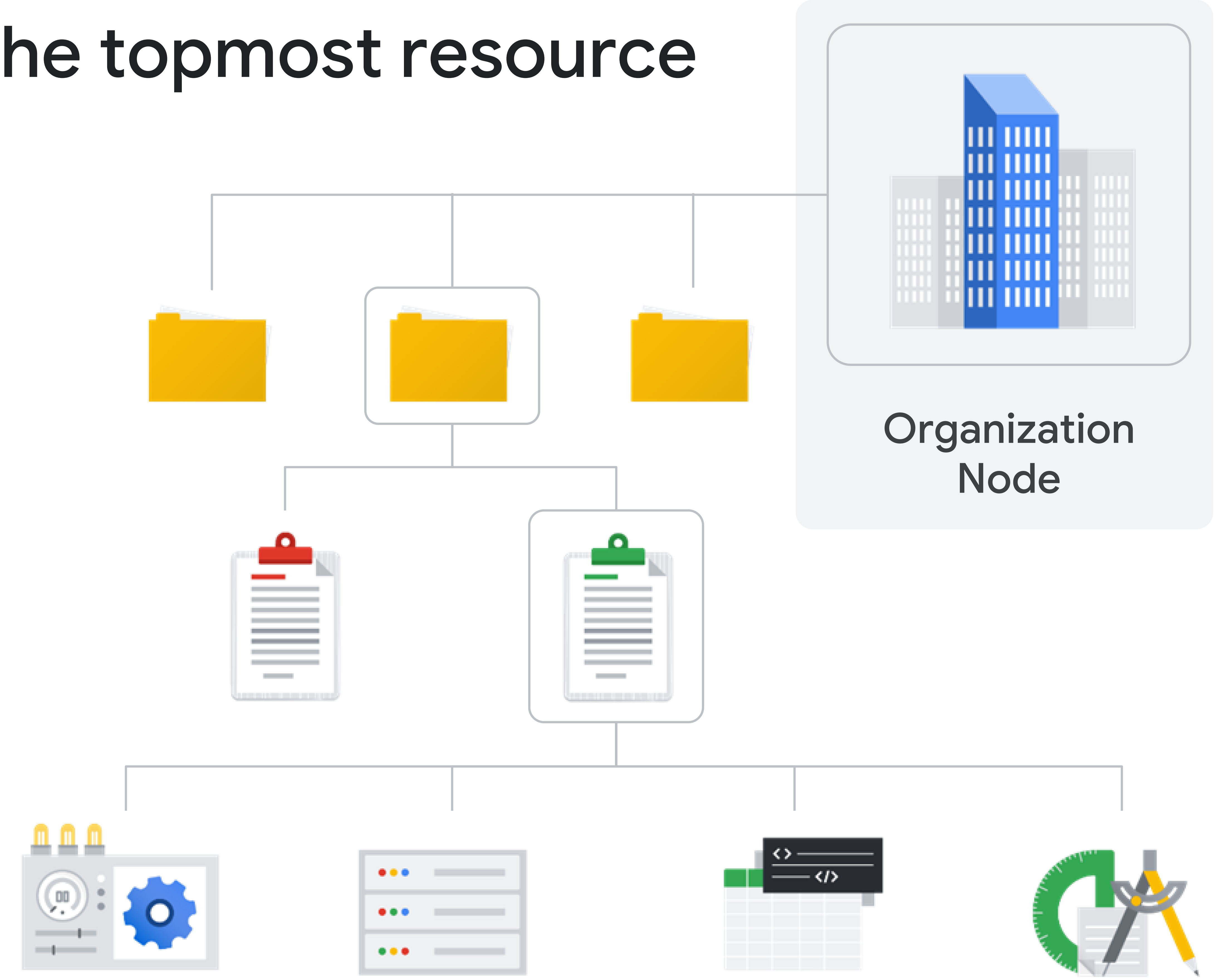
Folders group projects



Folders allow you
to **group resources** on
a per-department basis

Organization node is the topmost resource

Everything attached to the account goes under the organization node



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01 The Google Cloud console

02 Understanding projects

03 **Google Cloud billing**

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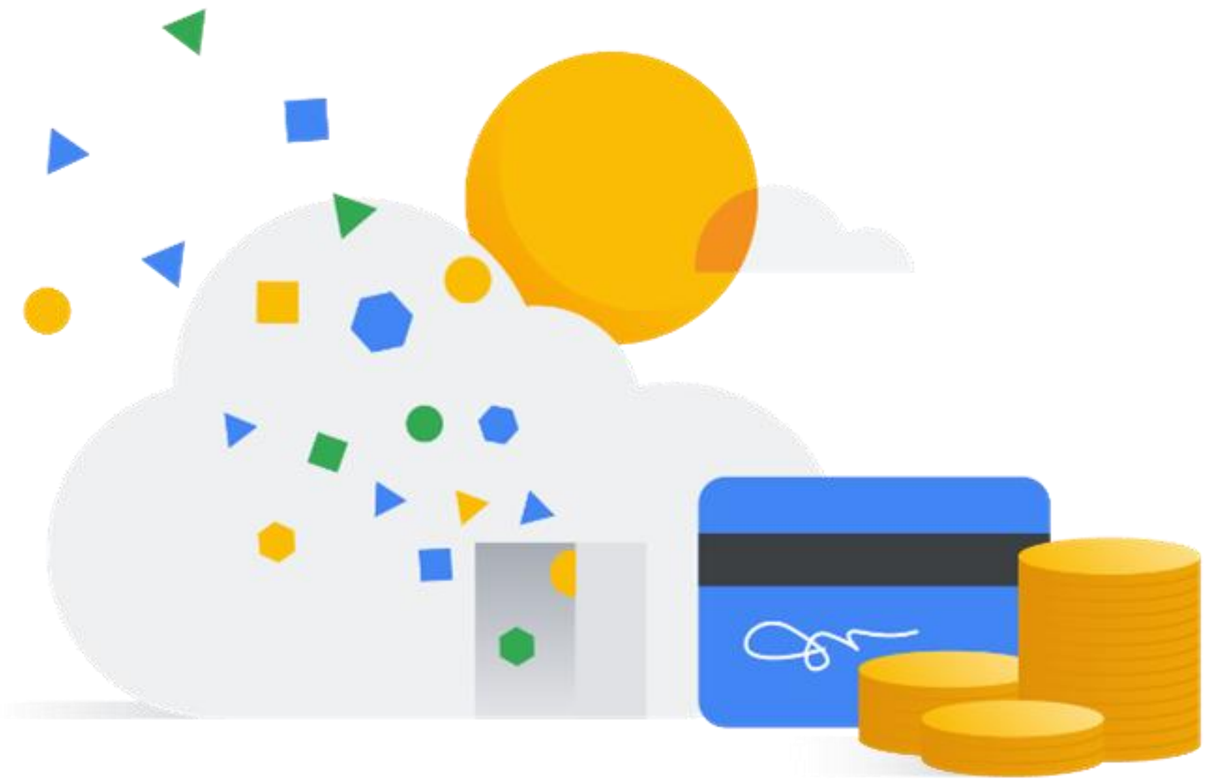
How billing works



Billing is established at the project level.

A billing account can be linked to zero or more projects.

Billing accounts are charged automatically and invoiced every month, or at every threshold limit.

Billing sub accounts can be used to separate billing by project.

Billing tools help to budget and monitor usage

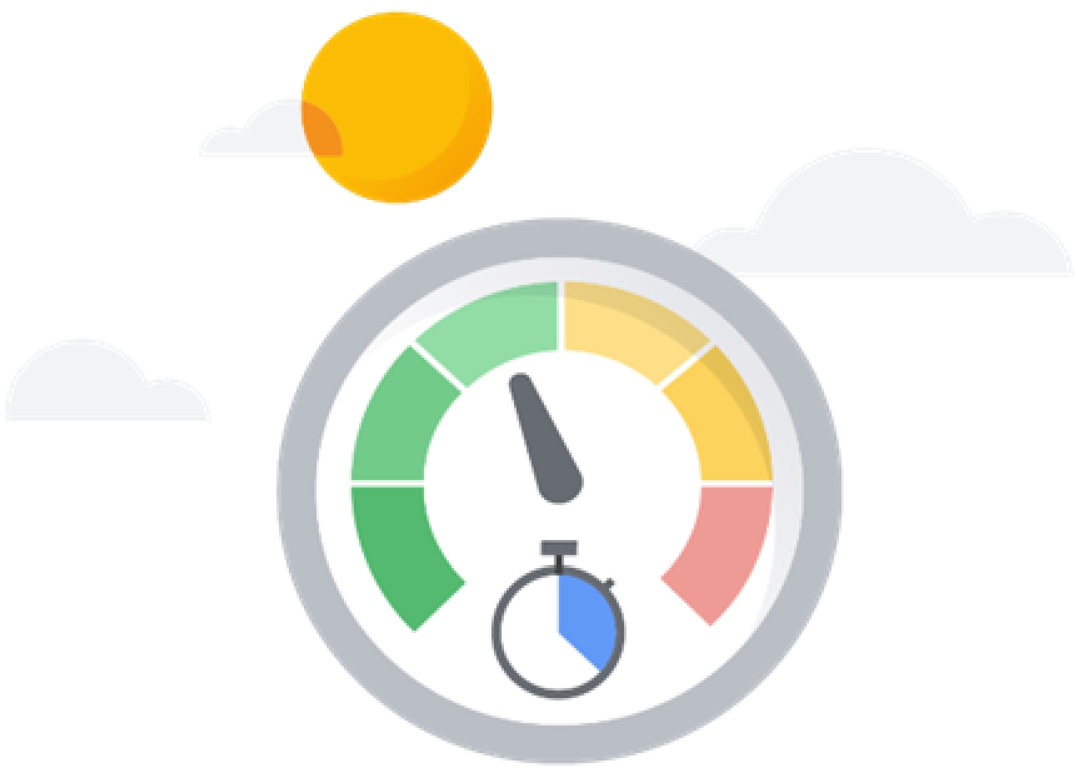
Budgets

Alerts

Reports

Quotas

Quotas are allocated at project-level

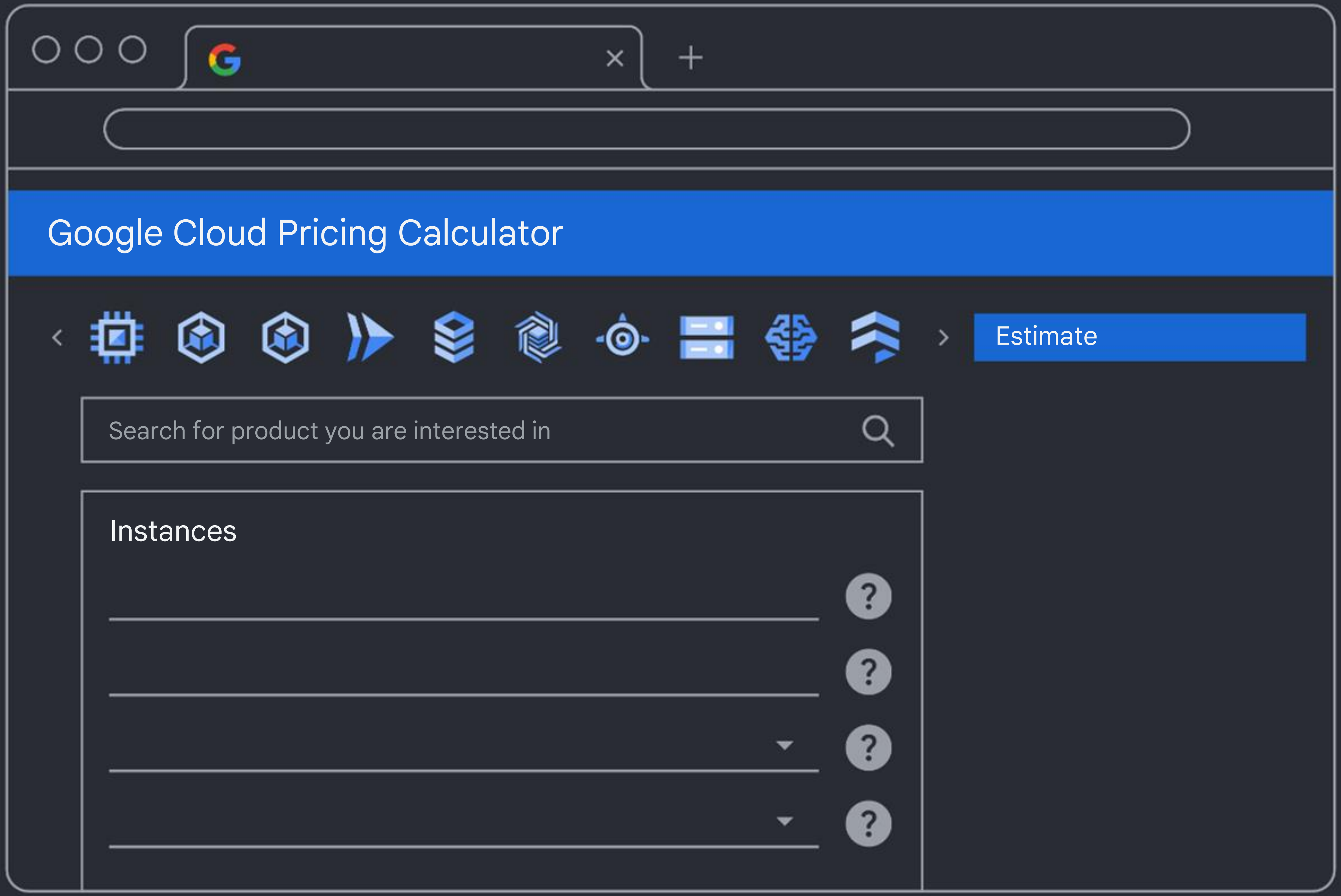


Rate quota

Resets after
a specific time

Allocation quota

Governs the number
of resources in a project



cloud.google.com/products/calculator

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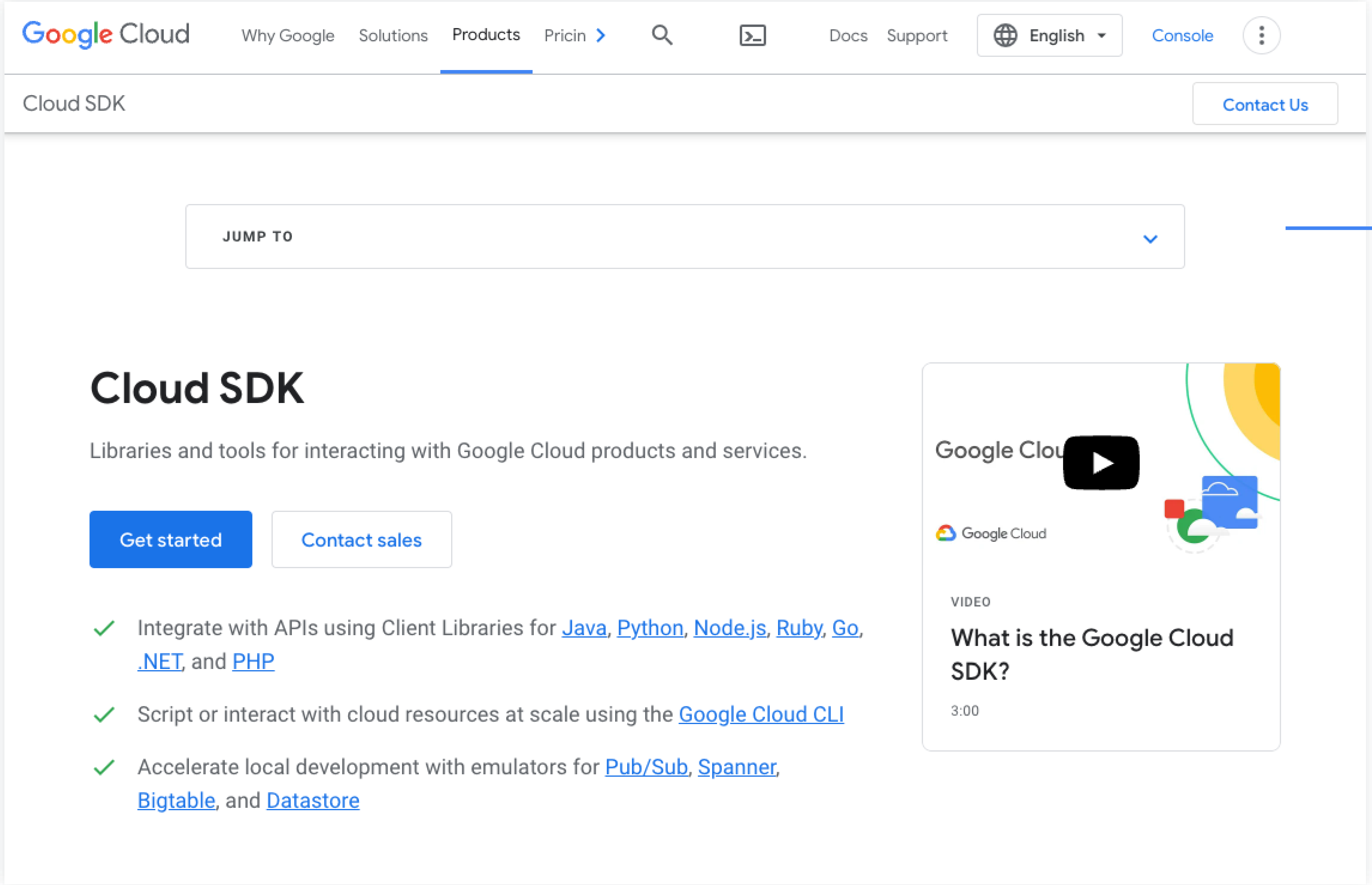
Cloud SDK is a collection of command line tools

- Set of tools to manage resources and applications hosted on Google Cloud
- Includes:
  - gcloud CLI - Provides the main command-line interface for Google Cloud products and services
  - gsutil - Provides access to Cloud Storage from the command line
  - bq - A command-line tool for BigQuery

Install the Cloud SDK



cloud.google.com/sdk

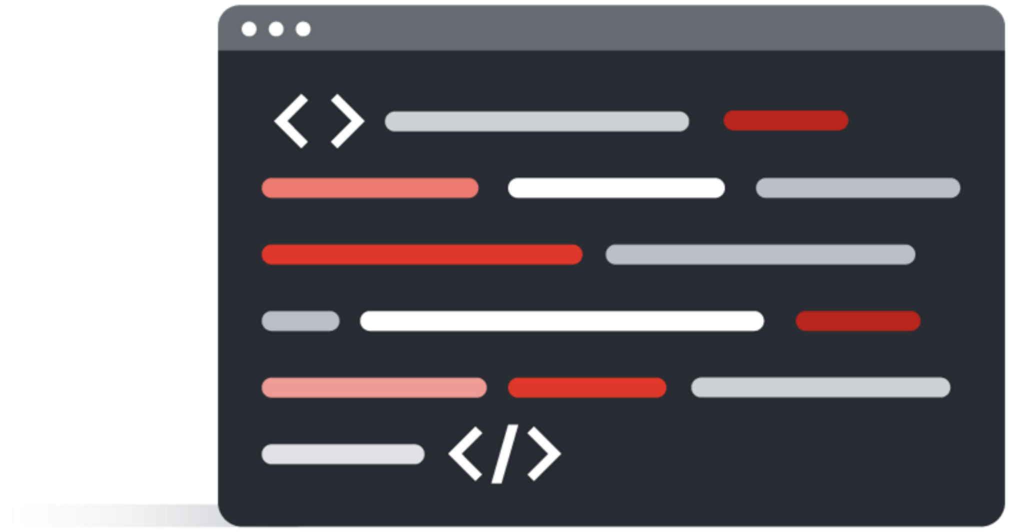
Configure the Cloud SDK

Google Cloud SDK Shell

```
C:\Program Files (x86)\Google\Cloud SDK: gcloud init
```

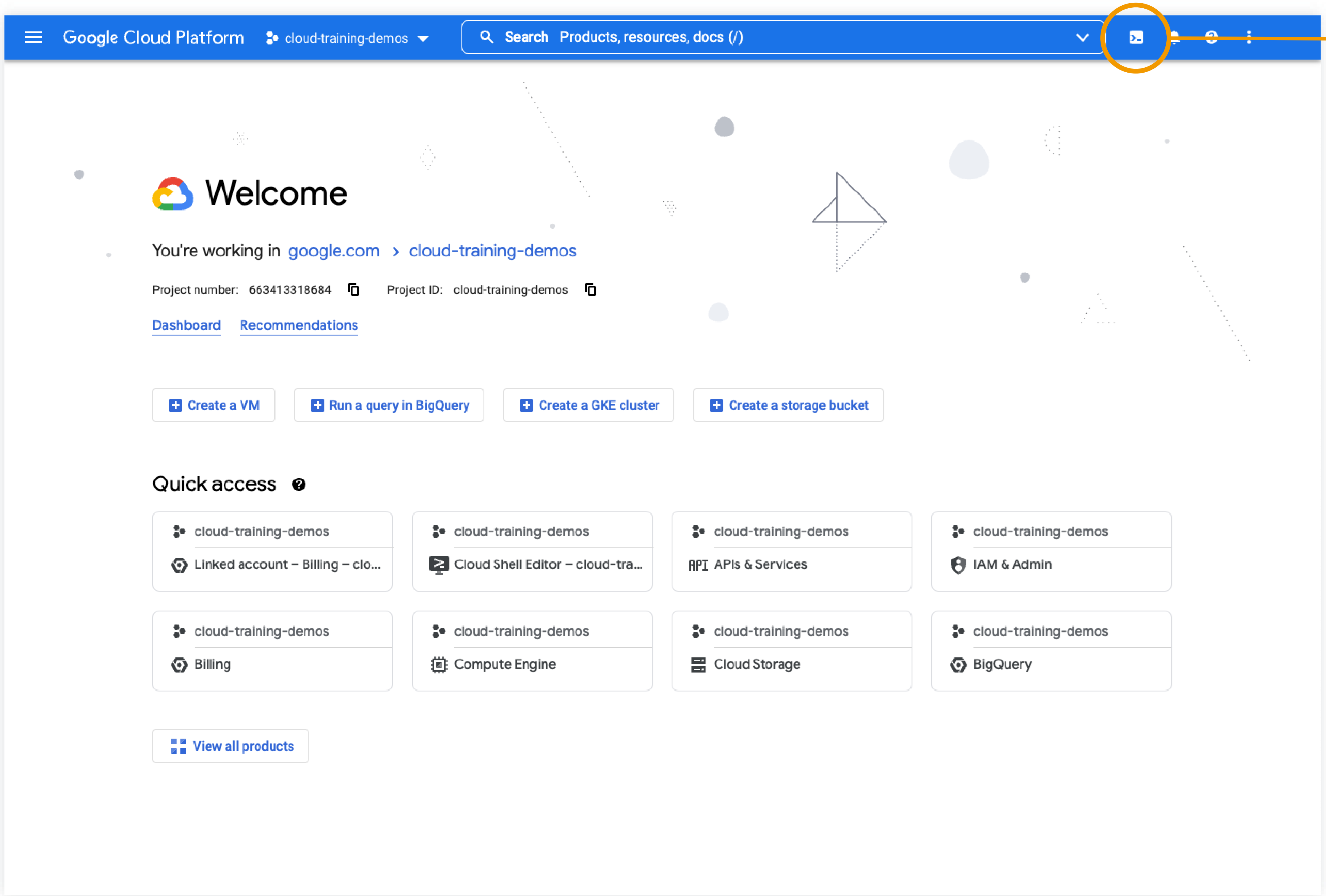
Agenda

Cloud Shell provides command line access to resources



- Provides command-line access to cloud resources directly from a browser
- Debian-based virtual machine with a persistent 5-GB home directory
- The Cloud SDK gcloud command and other utilities are always installed, available, up to date, and fully authenticated

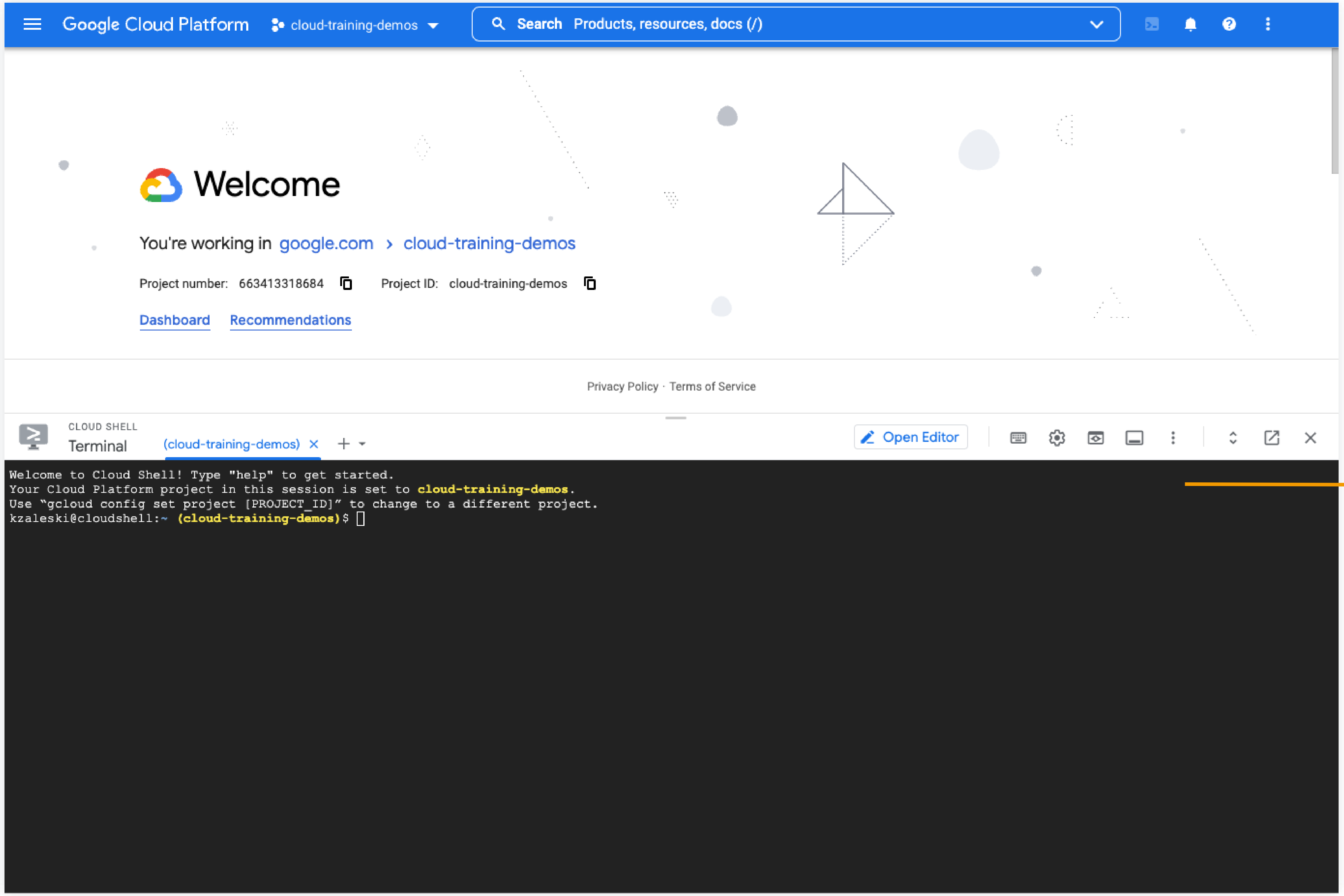
Starting Cloud Shell



console.cloud.google.com

Activate Cloud Shell

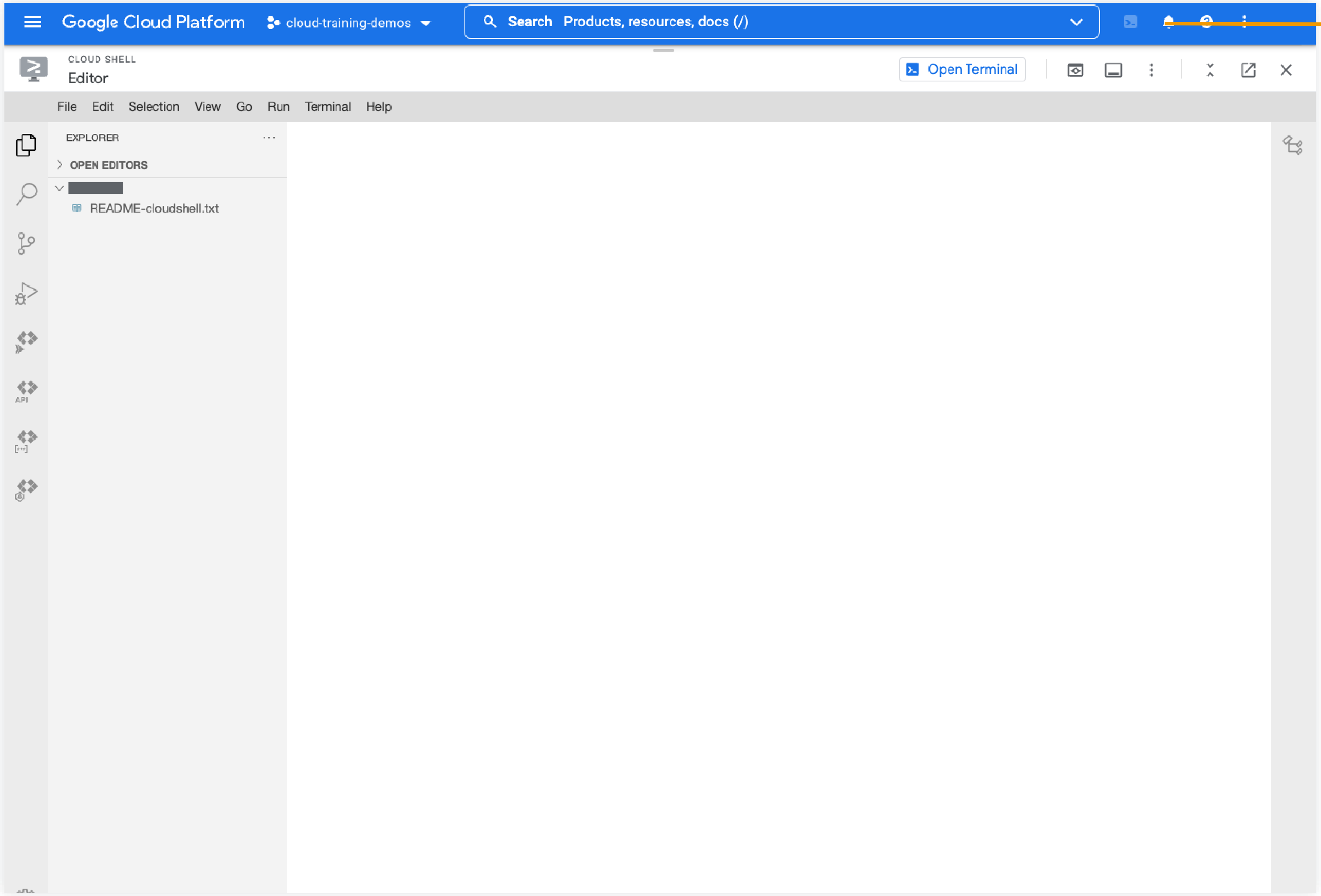
The Google Cloud Console and Cloud Shell



console.cloud.google.com

Cloud Shell terminal

The Cloud Shell code editor



console.cloud.google.com

Cloud Shell code editor

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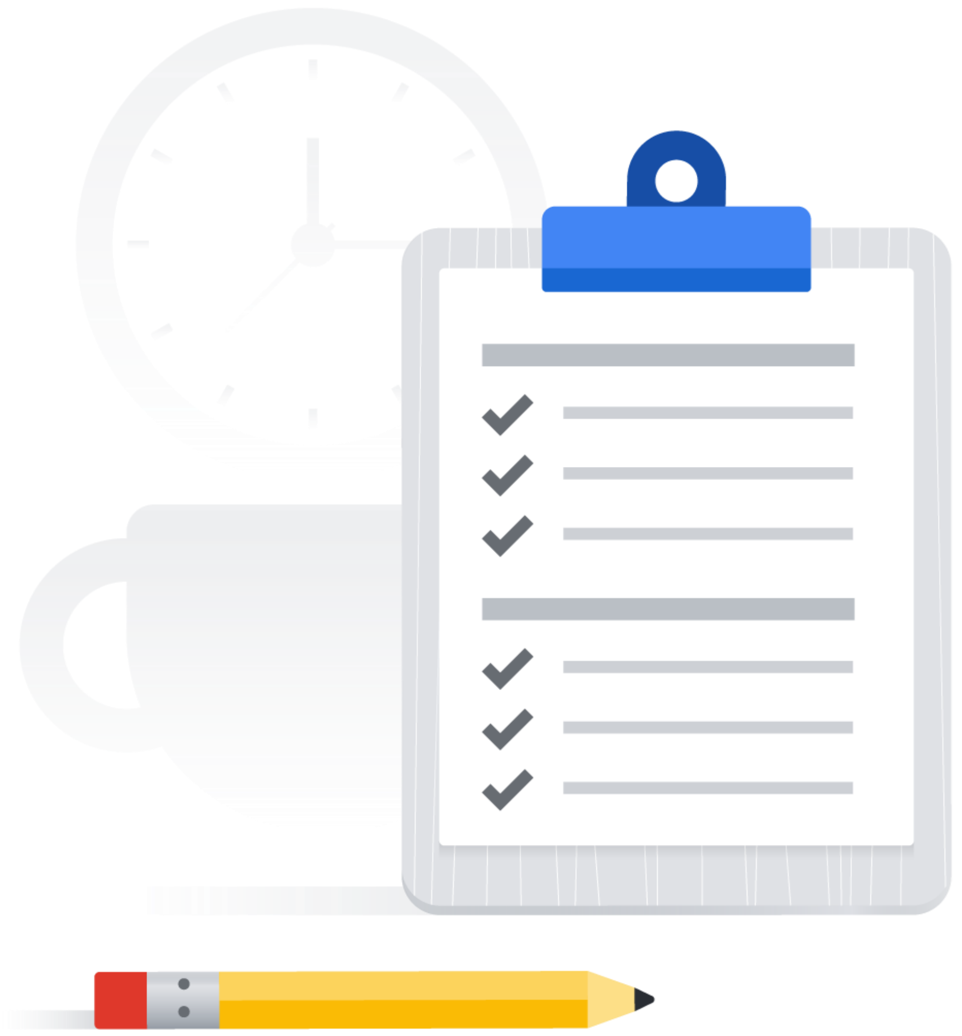
Lab: A Tour of Google Cloud Hands-on Labs

Hands-on lab



During this lab, you'll:

1. Get an introduction to the Qwiklabs platform and be able to key features of a lab environment.
2. Access the Google Cloud console with specific credentials.

Agenda

Lab: Getting Started with Cloud Shell and gcloud

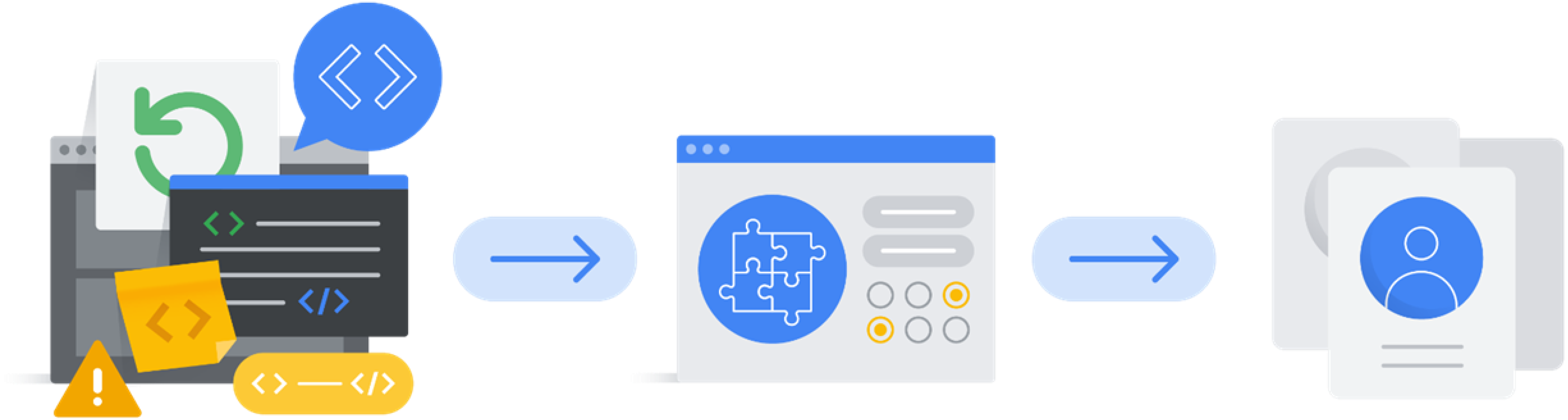
Hands-on lab

During this lab, you'll:

1. Get practice using gcloud commands.
2. Connect to storage services hosted on Google Cloud.

Agenda

Application Programming Interfaces (APIs)



Complex, changeable implementation

API
Simple, versioned interface

Users

APIs allow code to control your Cloud resources

- Google Cloud services offer APIs that allow code to be written to control them
- The Google APIs Explorer shows what APIs are available, and in what versions
- Google provides Cloud Client and Google API Client libraries
    - Languages currently represented: Java, Python, PHP, C#, Go, Node.js, Ruby and C++

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Manage your resources with the Cloud Mobile App

cloud.google.com/console-app

- Start, stop, and use SSH to connect into Compute Engine instances, and see logs
- Stop and start Cloud SQL instances
- Administer applications deployed on App Engine
- Up-to-date billing information for projects and alerts for those going over budget
- Customizable graphs showing key metrics
- Alerts and incident management

Agenda

Knowledge check

In the Google Cloud resource hierarchy, into which entity are resources organized?

A. Organization node

B. Folders

C. Projects

D. Zones

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Knowledge check

Which project identifier does not need to be globally unique?

A. Project ID

B. Project name

C. Project number

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Knowledge check

Which billing tool is designed to prevent the over-consumption of resources due to an error or a malicious attack?

A. Budgets

B. Alerts

C. Reports

D. Quotas

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Knowledge check

Which command line tool is part of the Cloud SDK?

A. bq

B. Git

C. Bash

D. SSH

Knowledge check

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Knowledge check

What is the purpose of APIs offered by various Google Cloud services?

A. APIs allow physical access to data centers.

B. APIs provide Google Cloud console access through a simple web-based graphical user interface.

C. APIs provide monthly pricing discounts.

D. APIs allow code to be written to control Google Cloud services.

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Summary

- Explored the Google Cloud console.
- Examined how projects are the basis for enabling and using Google Cloud services.
- Saw different billing options available in Google Cloud.
- Learned how to Install and configure the Cloud SDK.
- Explored the different use cases for using Cloud Shell and the Cloud Shell code editor.
- Learned how APIs work.
- Saw how to manage Google Cloud services from a mobile device.